HOMOTOPY LIMIT AND COLIMIT

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Abstract. These notes are for the talk at MCM homotopy reading seminar organized by Shizhang Li, Zhiyou Wu, Weizhe Zheng and Foling Zou during 2024 fall. We introduce the notation of homotopy limit and colimit, which is a remediation for the poor behavior of the homotopy category under universal constructions. We follow Vogt's original method to solve this coherence problem and we will also introduce the other viewpoint: the model category approach.

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Introduction

The universal construction: limits and colimits are of fundamental importance in categorical theory, probably the soul and core of the whole theory. In an ordinary category, they are defined to satisfy a characterized universal property. It becomes nasty when homotopy theory joints in. Classically, though the category of spaces $\mathcal{S}$ admits limits and colimits, the same fails for the homotopy category $\mathrm{h}\mathcal{S}$, and these limits and colimits fail to capture higher coherence issue even when they exist.

Example 0.1. Consider the inclusion of n-dimensional sphere $S^n \hookrightarrow D^n$, we have a map between diagrams of spaces:

$$\begin{array}{ccccc} D^n & \longleftarrow & S^n & \hookrightarrow & D^n \\ \downarrow & & \| & & \downarrow \\ * & \longleftarrow & S^n & \longrightarrow & * \end{array}$$

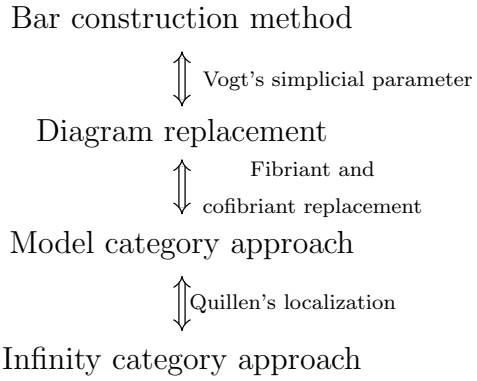
where each vertical arrow is a homotopy equivalence. However, the pushout of the upper diagram is S^{n+1}, while the pushout of the lower diagram is simply a point. They are not equivalent.

In order to solve the issue, we need to adjust the definition of "limits and colimits" to make them compatible with homotopy and retains the higher coherence data. The intuition is either we need to find some canonical choices amount these ordinary limits and colimits,

or we need to find another universal property to characterize them. Historically, there are several solutions:

- We mechanically define the homotopy limit and colimit by the ordinary limit and colimit of a huge diagram, roughly from a huge bar construction, which is "large enough" to capture all the data that are missing in the homotopy category. This is the method of Segal, Bousfield and Kan [Seg68], [BK72].
- We replace a limit or colimit diagram by a better one and define homotopy limit and colimit on this replacement. This is the method of Boardman and Vogt [Vog73].
- We regard homotopy limits and colimits as derived functors between homotopy categories of model categories. This is the viewpoint of Quillen [Qui06].
- We choose to live in the homotopy type-enriched categories, that is, the $(\infty, 1)$-category. We define all the universal constructions up to weak equivalences. This is the modern ∞-categorical viewpoint. The most adopted model is due to Lurie [Lur09].

All these methods are equivalent at least on the level of spaces. For instance, by Vogt's simplicial parameter argument, the first two methods are identified. The second and third methods are identical simply because of the identification of diagram replacement with the fibriant/cofibriant replacement.



The goal of this talk is to explain Vogt's original idea of the construction of the homotopy limits and colimits and how it foretells the forthcoming of Quillen's model category and quasi-category model of ∞-category. We will also introduce homotopy limits and colimits as derived functors between model categories. By duality, in this talk we will only focus on homotopy colimits.

1. Vogt's diagram method

In this section we will introduce Vogt's diagram method for homotopy colimits. Let $\mathcal{S}$ denote the category of compactly generated, weak Hausdorff spaces and let $\mathcal{S}_*$ be its pointed category. In classical homotopy theory, in order to describe higher coherence, one always demands some model spaces to "encode" these coherence. Examples include Stasheff's $\mathbb{A}_n$-spaces [Sta63], May's little cube $\mathbb{E}_n$- operads [May72], and Vogt's treatment to the *homotopy diagrams*.

1.1. **Homotopy diagrams.** Let $\mathcal{C}$ be a topological category, that is, a category that is enriched over $\mathcal{S}$. We denote the category of small topological categories by $\mathcal{S}$Cat. We use I to denote the unit interval $[0, 1]$.

Construction 1.1. We construct a functor $T\colon \mathcal{S}\mathrm{Cat} \to \mathcal{S}\mathrm{Cat}$ as follows: for $A, B \in \mathrm{obj}(\mathcal{C})$ we define

$$\mathcal{C}_n(A,B) = \{(f_n, \cdots, f_1) \in \mathrm{mor}(\mathcal{C})^n | f_n \circ \cdots \circ f_1 \colon A \to B\}$$

which is topologized as the subspace of $\mathrm{mor}(\mathcal{C})^n$. By convention, we put

$$\mathcal{C}_0(A,B) = \begin{cases} \{\mathrm{id}_A\} & \text{if } A = B \\ \emptyset & \text{otherwise} \end{cases}$$

We define $\mathrm{obj}(T\mathcal{C}) = \mathrm{obj}(\mathcal{C})$ and

$$T\mathcal{C}(A,B) := \coprod_{n \geq 0} \mathcal{C}_{n+1}(A,B) \times I^n$$

The composition is given by

$$(f_n, t_n, \cdots, t_1, f_0) \circ (g_m, s_m, \cdots, s_1, g_0) = (f_n, t_n, \cdots, f_0, 0, g_m, \cdots, g_0)$$

For any continuous functor $F\colon \mathcal{C} \to \mathcal{D}$, we have

$$TF(f_n, t_n, \cdots, f_0) = T(Ff_n, t_n, \cdots, Ff_0)$$

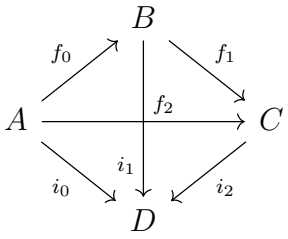
Note that in an obvious way there exists an augmented functor $\varepsilon\colon T\mathcal{C} \to \mathcal{C}$ and an inclusion functor $\eta\colon \mathcal{C} \to T\mathcal{C}$.

Definition 1.2. Let $\mathcal{C}$ be a topological category. A $\mathcal{C}$-diagram is an $\mathcal{S}$-enriched functor $F\colon \mathcal{C} \to \mathcal{S}_*$. A *homotopy* $\mathcal{C}$-diagram (h$\mathcal{C}$-diagram) is a $T\mathcal{C}$-diagram $D\colon T\mathcal{C} \to \mathcal{S}_*$ such that:

(1) $D(f_n, t_n, \cdots, f_0) = D(f_n, t_n, \cdots, f_{i+1}, t_{i+1}t_i, f_{i-1}, \cdots, f_0)$ for every $0 \leq i \leq n$ such that $f_i = \mathrm{id}$.
(2) $D(f_n, t_n, \cdots, f_0) = D(f_n, t_n, \cdots, t_{i+1}, f_i f_{i-1}, t_{i-1}, \cdots, f_0)$ for every $0 \leq i \leq n$ such that $t_i = 1$.

We replace a diagram $\mathcal{C} \to \mathcal{S}_*$ by another diagram $T\mathcal{C} \to \mathcal{S}_*$. The advantage of this replacement is it retains the homotopy coherence data. Suppose we are given a homotopy diagram D, one can think of each $D(f_i)$ as an arrow in the diagram, and each t_i as a parameter of higher homotopy.

Example 1.3. Consider $\mathcal{C}$ to be the following discrete category.



The homotopy diagram has the same vertex and arrows, with additional specified homotopies filling in each triangle and a higher homotopy filling in the tetrahedron.

Our next goal is to define morphisms between these homotopy diagrams. However, we will not only need these morphisms, but also homotopies and higher homotopies. In order to do this, we need some simplicial notations.

Notation 1.4. We let Δ_n denote the n-dimensional simplex: the poset $(\{0, \cdots, n\}, \leq)$. We let $\sigma_n^i\colon \Delta_{n+1} \to \Delta_n$ and $\delta_n^i\colon \Delta_{n-1} \to \Delta_n$ the coface and codegeneracy maps.

Definition 1.5. Let $\mathcal{C}$ be a topological category. We define a simplicial set $\mathscr{T}\mathcal{C}_\bullet$ as follows: For every $n \geq 0$, $\mathscr{T}\mathcal{C}_n$ is the (large) set of $\mathrm{h}(\mathcal{C} \times \Delta_n)$-diagrams. The face and degeneracy maps are given by

$$d_n^i(D) := T(D \circ (\mathrm{id} \times \delta_{\mathrm{n}}^{\mathrm{i}})) \quad s_n^i(D) := T(D \circ (\mathrm{id} \times \sigma_{\mathrm{n}}^{\mathrm{i}}))$$

It is somehow surprising that, this defines a quasi-category. Therefore, one can consider the morphisms, homotopies and higher homotopies of these homotopy diagrams. Let us recall the definition of quasi-categories.

Definition 1.6. Let $X \in \mathrm{sSet}$. X is called a *quasi-category* if for any n and $0 < i < n$, we can always solve the following lifting problem:

$$\begin{array}{ccc} \Lambda_n^i & \longrightarrow & X \\ \downarrow & \nearrow & \\ \Delta_n^i & & \end{array}$$

Here $\Lambda_n^i \subset \Delta_n$ denotes the i-th horn, obtained from the simplex Δ_n by deleting the interior and the face opposite the i-th vertex.

Theorem 1.7. *Let $\mathcal{C}$ be a small topological category. Then $\mathscr{T}\mathcal{C}_\bullet$ is a quasi-category.*

Sketch of proof. For notation simplicity, we denote $\mathcal{E} := \mathcal{C} \times \Delta_n$. We let $\mathscr{V}_n \subset T(\mathcal{C} \times \Delta_n) := T\mathcal{E}$ be the subcategory generated by the horn Λ_n^i, that is, the images of the face maps s_n^i for $0 \leq i \leq n$. We need to extend any V_n-diagram to a $T\mathcal{E}$-diagram, which is equivalent to extend the diagram map

$$D \colon \mathscr{V}(A, B) \times D(A) \to D(B)$$

to a map

$$D' \colon T\mathcal{C}(A, B) = \Big(\coprod_n \mathcal{C}_{n+1}(A, B) \times I^n \Big) \times D(A) \to D(B)$$

We do this by induction on n. We let $\mathcal{R}_n(A, B)$ denote $\mathcal{C}_{n+1}(A, B) \times I^n$, let $\mathcal{Q}_n(A, B) \subset \mathcal{R}_n(A, B)$ denote the subspace consisting of those $(f_n, t_n, \cdots, f_0)$ such that there exists some $t_i = 0$ or 1, or some $f_i = \mathrm{id}$. We let $\mathscr{V}_n(A, B) := \mathcal{R}_n(A, B) \cap \mathscr{V}(A, B)$. Suppose we have constructed a map

$$D'_k \colon \mathcal{R}_k(A, B) \times D(A) \to D(B)$$

that satisfies the condition of Definition 1.2. Then automatically we obtain a map:

$$D''_k \colon \big(\mathcal{Q}_{k+1}(A, B) \cup \mathscr{V}_{k+1}(A, B) \big) \times D(A) \cup \mathcal{R}_k(A, B) \times \{*\} \to D(B)$$

But one can show that this source is a neighborhood deformation retract of $\mathcal{R}_{k+1}(A, B)$. Thus we can always extend it to D_{k+1}. □

In order to move on, we will need some notations.

Definition 1.8. Let $\mathcal{C}$ be a topological small category.

(1) Let D_0, D_1 be two $\mathrm{h}\mathcal{C}$-diagrams. A *homotopy morphism* (h-morphism for short) from D_0 to D_1 is an $\mathrm{h}(\mathcal{C} \times \Delta_1)$-diagram H such that $d_0 H = D_0$ and $d_1 H = D_1$.
(2) Let E_0, E_1 be two $\mathcal{C}$-diagrams. A *morphism* from E_0 to E_1 is a natural transformation between them. For two morphism $f, g \colon E_0 \to E_1$, a (termwise) *homotopy* between f and g is a space-level homotopy between these maps.

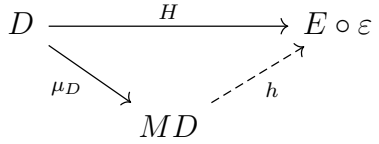
(3) We call a $\mathrm{h}(\mathcal{C}\times\Delta_n)$-diagram K *target reduced* if we have
$$K(f_j, t_j\cdots, f_0) = K(f_j\circ\cdots\circ f_i, t_i, \cdots, f_0)$$
once f_i lies in the last vertex of Δ_n. Let D be a $\mathrm{h}\mathcal{C}$-diagram and E be a $\mathcal{C}$-diagram. We call a h-morphism $H\colon D\to E\circ\varepsilon$ target reduced if it is a target reduced $\mathrm{h}(\mathcal{C}\times\Delta_1)$-diagram.

It follows directly that any h-morphism between $\mathrm{h}\mathcal{C}$-diagrams forgets to a morphism between $T\mathcal{C}$-diagrams.

1.2. **Homotopy colimits.** We're now ready to introduce the notation of homotopy colimits via homotopy diagrams. The definition of homotopy limits follows from a totally dual fashion. We begin with the diagram replacement functor.

Theorem 1.9. *Let $\mathcal{C}$ be a small topological category. Let D be a $\mathrm{h}\mathcal{C}$-diagram. Then there exists a $\mathcal{C}$-diagram MD and a target reduced map $\mu_D\colon D\to MD\circ\varepsilon$, such that:*

(1) Each underlying map $D(A)\to MD\circ\varepsilon(A)$ is a space-level trivial cofibration.
(2) For any $\mathcal{C}$-diagram E and any target reduced morphism $\tau\colon D\to E\circ\varepsilon$, we can always uniquely factor τ through μ_D.



Moreover, suppose that two target reduced maps $H, H'\colon D\to E\circ\varepsilon$ are simplicially homotopic, then the induced morphisms h, h' are termwise homotopic.

Proof. We can directly construct MD. We define:
$$\begin{aligned} MD(A) :=& \mathrm{colim}_{B\in\mathcal{C}_{/A}} T\mathcal{C}(B,A)\times D(B)\\ =& \coprod_n \coprod_{B\in\mathcal{C}} \mathcal{C}_{n+1}(B,A)\times I^n\times D(B)/\sim \end{aligned}$$
with the relations given by
$$(f_n, t_n, \cdots, f_0)\sim \begin{cases} (f_n,\cdots, f_{i+1}, t_{i+1}t_i, f_{i-1},\cdots, f_0) & \text{if } f_i = \mathrm{id}\\ (f_n,\cdots, t_{i+1}, f_if_{i-1}, t_{i-1},\cdots, f_0) & \text{if } t_i = 1\\ (f_n,\cdots, f_i, D(f_{i-1}, t_{i-1},\cdots, f_0)) & \text{if } t_i = 0 \end{cases}$$
The first claim can be checked directly since we already have a coordinate representation. Suppose we are given a target reduced morphism $\tau\colon D\to E\circ\varepsilon$, then we can write down the morphism $h\colon MD\to E\circ\epsilon$ as
$$h(f_n, t_n, \cdots, f_0) = \tau(f_n, 0, t_n, f_{n-1}, \cdots, f_0)$$
The property follows directly. □

We are going to introduce some formal notations. We let $\mathscr{H}\mathcal{C}$ denote the category of $\mathrm{h}\mathcal{C}$-diagrams and simplicial homotopy classes of morphisms. Let $\mathscr{K}\mathcal{C}$ denote the category of $\mathcal{C}$-diagrams and termwise homotopy classes of morphisms. Let $J\colon\mathscr{K}\mathcal{C}\to\mathscr{H}\mathcal{C}$ be the augmented functor sending E to $E\circ\varepsilon$.

Proposition 1.10. *We have an adjunction*

$$M\colon \mathscr{H}\mathcal{C} \rightleftharpoons \mathscr{K}\mathcal{C} : J$$

where the left adjoint is fully faithful.

Proof. Let $\mathscr{T}(D, E)$ denote the space of target-reduced morphisms between D and $E \circ \varepsilon$. Then it follows from the definition that we have

$$\mathscr{H}\mathcal{C}(D, JE) \simeq \mathscr{T}(D, E) \simeq \mathscr{K}\mathcal{C}(MD, E)$$

Therefore we obtain the adjunction. Moreover, the unit map $\mathrm{id} \to JM$ is given by μ_D, which is an equivalence by Theorem 1.9. □

We can now define what is a homotopy colimit of a diagram.

Definition 1.11. The homotopy colimit functor is defined to be the composite

$$\mathrm{hocolim} := \mathrm{colim} \circ M\colon \mathscr{H}\mathcal{C} \to \mathcal{S}_*$$

As a consequence of Theorem 1.9, we have

$$\coprod_n \coprod_{A,B\in\mathcal{C}} \mathcal{C}_{n+1}(B, A) \times I^n \times D(B)/\sim$$

It follows immediately from the above lemma and classical consequences that hocolim is a left adjoint of the constant diagram functor.

We end this section by some examples.

Example 1.12. Consider the diagrams in the category of pointed spaces $\mathcal{S}_*$.

(1) Let $D = Y \longleftarrow X \longrightarrow Z$, the homotopy colimit of D is the mapping cylinder.
(2) Let $D = X \rightrightarrows Y$, the homotopy colimit of D is the mapping torus.

2. Quillen's model category approach

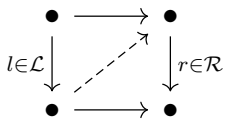
Vogt's explanation of homotopy colimits and limits through diagram replacement is somehow complicated. Quillen concluded this in a more formal language, which summarized a general pattern when homotopy theory and categorical theory meet. That is, the model category approach.

2.1. A quick introduction to model category.

We quickly introduce model category theories. For a detailed introduction, see for example [Hov07].

Definition 2.1. Let $\mathcal{C}$ be a category. A weak factorization system $(\mathcal{L}, \mathcal{R})$ is comprised of two classes of morphisms $\mathcal{L}$ and $\mathcal{R}$ such that:

(1) Every morphism in $\mathcal{C}$ can be factored as a morphism in $\mathcal{L}$ followed by a morphism in $\mathcal{R}$.
(2) The maps in $\mathcal{L}$ have the left lifting property with respect to each map in $\mathcal{R}$ and the maps in $\mathcal{R}$ have the right lifting property with respect to each map in $\mathcal{L}$. That is, any commutative square



admits a diagonal filler making both triangle commute.
(3) The classes $\mathcal{L}$ and $\mathcal{R}$ are closed under retractions in the arrow category.

Definition 2.2. A model structure on a category $\mathcal{C}$ consists of three classes of maps: the weak equivalences $\mathcal{W}$, cofibrations cofib and fibrations fib. They should satisfy:

(1) $\mathcal{W}$ satisfies 2-of-3.
(2) $(\text{cofib}, \text{fib} \cap \mathcal{W})$ and $(\text{cofib} \cap \mathcal{W}, \text{fib})$ define two weak factorization systems.

Remark 2.3. In general, we will assume that the factorization system is functorial although for most purposes the above axioms are already sufficient.

Now we assume that $\mathcal{C}$ admits initial and terminal objects.

Definition 2.4. We say that an object $X \in \mathcal{C}$ is *fibriant* if the map $X \to *$ is a fibration. We say X is *cofibriant* if the map $\emptyset \to X$ is a cofibration.

Definition 2.5. A *fibriant replacement* of X is a factorization $X \xrightarrow{\sim} X_{\text{fib}} \to *$ such that X_{fib} is fibriant. A *cofibriant replacement* of X is a factorization $\emptyset \to X_{\text{cof}} \xrightarrow{\sim} X$ such that X_{cof} is cofibriant.

When assuming functorial factorization, the fibriant and cofibriant replacements define two functor

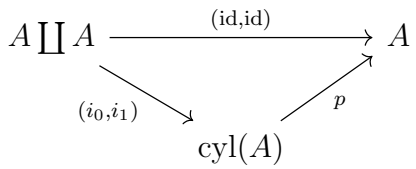
$$R \colon \mathcal{C} \to \mathcal{C} \quad Q \colon \mathcal{C} \to \mathcal{C}$$

such that for any object X, one can always find a zig-zag weak equivalence between RQX and QRX.

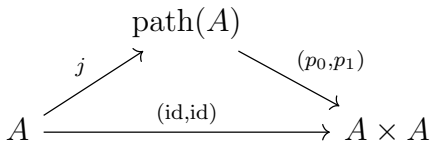
We can now define homotopy between morphisms in a model category.

Definition 2.6. Let A be an object in a model category $\mathcal{C}$, where $\mathcal{C}$ admits enough colimits.

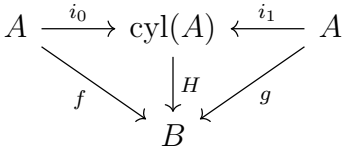
(1) A cylinder object for A is a cofibration-trivial fibration factorization:



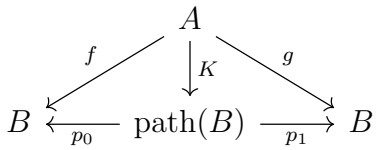
(2) Dually, a path object for A is a trivial cofibration-fibration factorization:



(3) Suppose we are given a pair of morphisms $f, g \colon A \to B$, a left homotopy between f and g is a morphism $H \colon \text{cyl}(A) \to B$ that fits into the diagram:



(4) Dually, a right homotopy between f and g is a morphism $K\colon A \to \mathrm{path}(B)$ that fits into the following diagram:



The concept of left and right homotopy can be organized into the concept of homotopy, provided we restrict ourselves on good objects.

Proposition 2.7. *Suppose that A is cofibriant and B is fibriant, then the equivalent relation on the morphism set defined by left homotopy coincide with that is defined by the right homotopy.*

2.2. **Homotopy limits and colimits as derived functors.** We now explain how to define homotopy limits and colimits as derived functors between model categories.

Definition 2.8. Suppose that $\mathcal{C}$ is a model category. We define the *homotopy category* of $\mathcal{C}$, denoted by $\mathrm{h}\mathcal{C}$ as follows:

(1) The objects of $\mathrm{h}\mathcal{C}$ are those cofibriant-fibriant objects.
(2) The morphisms are the homotopy classes of morphisms.

As a consequence, there is a forgetful functor $\pi\colon \mathcal{C} \to \mathrm{h}\mathcal{C}$ with is given by composing cofibriant-fibriant replacement with the projection.

Definition 2.9. Let $\mathcal{C}_1$, $\mathcal{C}_2$ be two model categories. Let $F\colon \mathcal{C}_1 \to \mathcal{C}_2$ be a functor.

(1) We say that F is *left Quillen* if it preserves cofibration, trivial cofibration and cofibriant objects.
(2) Dually, we say that F is *right Quillen* if it preserves fibration, trivial fibration and fibriant objects.
(3) Suppose that F is left Quillen. The *left derived functor* of F is a left Kan extension $\mathbb{L}F\colon \mathrm{h}\mathcal{C}_1 \to \mathrm{h}\mathcal{C}_2$:

$$\begin{array}{ccc} \mathcal{C}_1 & \xrightarrow{F} & \mathcal{C}_2 \\ {\scriptstyle \pi_1}\downarrow & & \downarrow{\scriptstyle \pi_2} \\ \mathrm{h}\mathcal{C}_1 & \xrightarrow[\mathbb{L}F]{} & \mathrm{h}\mathcal{C}_2 \end{array}$$

(4) Dually, suppose that F is right Quillen. The *right derived functor* of F is a right Kan extension $\mathbb{R}F\colon \mathrm{h}\mathcal{C}_1 \to \mathrm{h}\mathcal{C}_2$:

$$\begin{array}{ccc} \mathcal{C}_1 & \xrightarrow{F} & \mathcal{C}_2 \\ {\scriptstyle \pi_1}\downarrow & & \downarrow{\scriptstyle \pi_2} \\ \mathrm{h}\mathcal{C}_1 & \xrightarrow[\mathbb{R}F]{} & \mathrm{h}\mathcal{C}_2 \end{array}$$

Though the definition of derived functors are through Kan extension. To compute derived functors suffices to find cofibriant or fibriant replacement.

Theorem 2.10. *In our definition above, the left derived functor is given by $\mathbb{L}F = F \circ R$, the right derived functor is given by $\mathbb{R}F = F \circ Q$.*

Finally, we can write down the definition of homotopy limits and colimits.

Definition 2.11. Let $\mathcal{C}$ be a model category and I be a small category. There is a *projective model structure* on the diagram category $\mathcal{C}^I$.

(1) We define the homotopy colimit functor as the left derived functor of $\operatorname{colim}_I : \mathcal{C}^I \to \mathcal{C}$.
(2) Dually, we define the homotopy limit functor as the right derived functor of $\lim_I : \mathcal{C}^I \to \mathcal{C}$.

Example 2.12. Let us consider the diagram $* \leftarrow S^n \to *$ in $\mathcal{S}_*$. In order to compute the homotopy colimit (pushout) of this diagram, we need to apply colimit to the cofibriant replacement of this diagram, which is exactly $D^n \leftarrow S^n \to D^n$.

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